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Rosalind Z. Wiggins
YPFS, Yale School of Management

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Eurozone: Central Bank Swaps to Ireland and the United Kingdom, 2010¹

Rosalind Z. Wiggins^{2,3}

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Abstract

In 2010, Ireland was experiencing a bank crisis and a sovereign debt crisis as a result of fallout from the broader Global Financial Crisis that had begun in 2007. Irish banks were having difficulty raising funds on wholesale funding markets and had borrowed extensively from the European Central Bank (ECB) in euros and US dollars. Despite the Irish government's guaranteeing the Irish banks and taking other extraordinary actions, analysts and European Union (EU) and United Kingdom officials worried that Irish banks might still default, since the Irish banking sector had significant linkages with the UK and its banking sector. As a result, on December 10, 2010, the ECB and Central Bank of Ireland (CBI) entered into a bilateral, unidirectional swap arrangement pursuant to which the CBI could request pounds sterling (GBP) in exchange for euros. A week later, the ECB and the Bank of England (BoE) entered into a bilateral, unidirectional swap agreement pursuant to which the BoE would provide up to GBP 10 billion (USD 15.6 billion) to the ECB in exchange for euros to address any liquidity constraints. Under the terms of the BoE-ECB swap, the ECB could then provide the drawn pounds sterling to the CBI, which could lend the funds to banks in its jurisdiction. The EU and International Monetary Fund also provided a EUR 85 billion (USD 113 billion) aid package to Ireland. The ECB-CBI agreement was extended once and terminated on September 28, 2012. The BoE-ECB agreement was extended three times and expired on September 30, 2014; the CBI reported that one bank utilized the facility in 2010.

Keywords: euro system, foreign currency swap, Ireland, liquidity

¹ This case study is part of the Yale Program on Financial Stability (YPFS) selection of New Bagehot Project modules considering central bank swap line programs. Cases are available from the *Journal of Financial Crises* at <https://elischolar.library.yale.edu/journal-of-financial-crises/>.

² Director, the Global Financial Crisis Project, YPFS, and Lecturer, Yale School of Management.

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Overview

By 2010, the Global Financial Crisis, which had begun in 2007, had had significant impacts on the financial systems of the United States and Europe. The Irish economy was hit particularly hard as credit tightened and housing prices crashed. The economy suffered a 7.1% contraction in 2009, which caused a steep decline in tax revenue (Saltmarsh 2010). These circumstances caused significant stresses on Irish banks, and on September 30, 2008, the government issued a two-year blanket guarantee of all debts and liabilities of six major banks and financial institutions (Dow Jones News 2008; Murray-Brown and Dennis 2008; Reuters Staff 2008). Despite this unprecedented action, credit markets continued to tighten, and other banks became reluctant to lend to Irish banks (CBI 2011).

For this reason, the Irish banks borrowed euros heavily from European Central Bank (ECB) liquidity facilities (CBI 2011). As of November 2010, the ECB had lent EUR 140 billion (including emergency liquidity assistance) or approximately 85% of Irish GDP. This represented approximately 25% of the ECB's outstanding lending, to Irish banks, "an unprecedented level of exposure to a country such as Ireland, whose share in the capital of the ECB was less than one percent" (ECB 2010b).

During 2009 and 2010, Irish sovereign debt was downgraded several times by the major credit rating agencies. A media report regarding a July 2010 downgrade by Moody's Investors Service quoted a senior official saying, "Today's downgrade is primarily driven by the Irish government's gradual but significant loss of financial strength, as reflected by its deteriorating debt affordability" (Saltmarsh 2010). Continued difficulties, including the expiration of the

Key Terms

Purpose: ECB-CBI agreement—"to facilitate the function of financial markets and provide liquidity in GBP" (ECB and CBI 2010). BoE-ECB agreement—The facility "allows pounds sterling to be made available to the Central Bank of Ireland as a precautionary measure, for the purpose of meeting any temporary liquidity needs of the banking system in that currency" (ECB 2011b, 100)

Participating Parties	ECB, BoE, CBI
Type of Swap	ECB-CBI and BoE-ECB: Reciprocal and unidirectional
Currencies Involved	Pounds sterling and euros
Launch Date	ECB-CBI: Dec. 10, 2010; BoE-ECB: Dec. 17, 2010
End Date	ECB-CBI: Sept. 28, 2012; BoE-ECB: Sept. 30, 2014
Date of First Usage	The CBI reported in its 2010 annual report that the BoE-ECB swap line was used by one Irish institution, but no details were given
Interest Rate and Fees	Not available
Amount Authorized	ECB-CBI: Not stated; BoE-ECB: GBP 10 billion (USD 15.6 billion)
Peak Usage Amount and Date	Not available
Downstream Use/Application of Swap Funds	ECB-CBI: To support financial markets and provide liquidity in GBP to Irish financial markets; BoE-ECB: To enable the ECB to provide GBP to the CBI
Outcomes	ECB-CBI: Expired Sept. 2012; BoE-ECB: Expired Sept. 2014 <i>(continued)</i>

Irish government’s guarantee on bank debt in October 2010, resulted in a five-notch downgrade by Moody’s on December 17, 2010 (ECB 2010b; French 2023). Moody’s cited the following factors and highlighted the government’s interconnection with the struggling financial sector: “(1) the crystallization of bank related contingent liabilities, (2) the increased uncertainty regarding the country’s economic outlook; and (3) the decline in the Irish government’s financial strength” (Moody’s 2010). As a consequence, the Irish government lost access to funding markets (ECB 2010b). At one point, one lender haircut Irish sovereign securities by 80% of their face value before accepting them as collateral (Baudino, Murphy, and Svoronos 2020).	<table><tr><th data-bbox="818 191 1029 233">Notable Features</th><td data-bbox="1078 191 1433 573">The ECB required a “letter of comfort” from the Irish government prior to entering into the swap. The ECB announced that it would lend the pounds sterling drawn under the BoE-ECB swap line only to the CBI. This case is unique in that it concerns two swap agreements in a linear relationship among three central banks</td></tr></table>	Notable Features	The ECB required a “letter of comfort” from the Irish government prior to entering into the swap. The ECB announced that it would lend the pounds sterling drawn under the BoE-ECB swap line only to the CBI. This case is unique in that it concerns two swap agreements in a linear relationship among three central banks
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Ireland had long been a significant trading partner of the United Kingdom, which was a member of the European Union (EU) but not the eurozone. In discussions with the International Monetary Fund (IMF) and EU about the United Kingdom’s contribution to a bailout for Ireland, then-UK Chancellor of the Exchequer George Osborne commented that Ireland’s banking sector was “closely linked to the UK’s.”⁴ He stated, “We are doing this because it is overwhelmingly in Britain’s national interest that we have a stable Irish economy and banking system” (BBC News 2010b). In addition, some Irish banks were having difficulties funding their UK deposit books. The UK ultimately did provide a direct loan of 3.2 billion pounds sterling (GBP) to Ireland. The country also received a EUR 85 billion rescue package funded by the EU and IMF (Baudino, Murphy, and Svoronos 2020; BBC News 2010a; Neuger and Kennedy 2010). (See Key Design Decision No. 2, Part of a Package.)

On December 10, 2010, the ECB and the Central Bank of Ireland (CBI) entered into a temporary bilateral, unidirectional currency swap agreement pursuant to which the CBI could request pounds sterling in exchange for euros to manage financial markets and make liquidity in that currency available to financial markets. The agreement did not state any maximum amount that could be borrowed. Neither bank issued a press release announcing this agreement; the ECB noted it in its annual reports (ECB and CBI 2010).

On December 17, 2010, the Bank of England (BoE) also entered into a temporary bilateral, unidirectional currency swap agreement with the ECB pursuant to which the ECB would be able to make funding in pounds sterling available to Irish banks through the CBI (CBI 2011; Reuters 2010). The ECB announcement termed the agreement “a precautionary measure”

⁴ Ireland accounted for 5% of Britain’s total exports and two-fifths of Northern Ireland’s exports; two of the four largest High Street banks in Northern Ireland were Irish owned, accounting for almost one-quarter of personal accounts; and two Irish banks also issued sterling notes in Northern Ireland (BBC News 2010b; Osborne 2010). In addition, UK banks had GBP 150 billion in exposures to Ireland (McDonald and Treanor 2010).

for the purpose of meeting any temporary liquidity needs of the Irish banking system for pounds sterling (BoE 2010).

Under the 2010 BoE-ECB agreement, the BoE, if requested, would provide the ECB with up to GBP 10 billion⁵ in exchange for euros (BoE 2010). While the BoE announcement stated that the ECB could use drawn pounds sterling to lend to “its counterparties,” the ECB press release was more specific: “The agreement allows pounds sterling to be made available to the Central Bank of Ireland as a precautionary measure, for the purpose of meeting any temporary liquidity needs of the banking system in that currency” (ECB 2011b, 100, 100).

We were able to access the ECB-CBI swap agreement but not the BoE-ECB agreement. Our information regarding the latter is informed by secondary sources such as press releases and annual reports, and at times, we reference a 2014 swap agreement between the parties for comparison.

Although we do not know the specific details of the BoE-ECB agreement, in general, it would usually be up to the borrowing central bank, in this case, the ECB, to determine the use of drawn funds⁶ (BoE 2010; ECB 2010a) (see also Key Design Decision No. 6, Communication).

The intent of the ECB as described in its 2010 annual report seemed to be that once it drew pounds under the BoE-ECB agreement, it could lend those funds to the Central Bank of Ireland (pursuant to the ECB-CBI agreement, “as a precautionary measure, for the purpose of meeting any temporary liquidity needs of the banking system in that currency” (ECB 2011b, 100). These needs would presumably be met through the CBI lending pounds to financial markets and banks under operations it, not the ECB, would administer (ECB and CBI 2010). This would be a process similar to how the ECB generally managed liquidity arrangements throughout the euro area and would be consistent with its US dollar-providing and Swiss franc-providing operations—lending through national central banks (NCBs) rather than directly. (See Key Design Decision No. 5, Administration.)

The BoE-ECB swap line agreement had an original expiration date of September 30, 2011. It was renewed three times and expired on September 30, 2014 (ECB 2011a; ECB 2012; ECB 2013a). The Central Bank of Ireland reported in its 2010 annual report, with respect to this swap line, that “one credit institution has availed of this facility” as of May 20, 2011 (CBI 2011, 117).

Summary Evaluation

The ECB reported in 2014 that the BoE-ECB swap for Ireland “helped to ensure an orderly restructuring of the Irish banking system at a time when Irish banks were experiencing impaired access to the wholesale market and prevented a fire sale of assets and a too abrupt deleveraging process” (ECB 2014, 74).

⁵ According to the *Wall Street Journal*, GBP 1 = USD 1.56 on December 31, 2010.

⁶ It is interesting to note that a 2014 swap agreement between the BoE and ECB is silent as to any acceptable or restricted uses of drawn funds by the borrower (ECB and BoE 2014).

Although we have not located any articles that evaluate the efficacy of the 2010 ECB-CBI and 2010 BoE-ECB swaps independently, we have located a number of sources that comment on the rescue efforts in the aggregate. Thorhallsson and Kirby (2012) finds that, “in the immediate aftermath of the crisis, membership of the ECB was an advantage to the Irish authorities as they were able to draw on the Bank’s support” (Thorhallsson and Kirby 2012, 808). However, Thorhallsson and Kirby further discusses the controversial evolution of the Ireland–ECB relationship in light of the outsized liquidity needs of the Irish financial system during an unprecedented crisis and the ECB’s efforts to have the Irish accept an international support package (Thorhallsson and Kirby 2012, 808–9). The Bank for International Settlements (BIS) also concludes that the Irish financial authorities “misdiagnosed the nature of the banking crisis as one of liquidity, linked to global market turmoil and overreliance on wholesale funding, as opposed to one of solvency” (Baudino, Murphy, and Svoronos 2020, 1). In addition to continued liquidity provision, Ireland would eventually employ a government guarantee and asset management vehicle to restructure its financial system (Baudino, Murphy, and Svoronos 2020).

By 2013, the IMF reported that Ireland’s economy appeared to be making some gains toward stabilizing: ECB lending was down, and employment was up. However, “policy implementation remains on track yet in total, near-term growth prospects [were] weaker and significant fiscal, financial sector and unemployment challenges remain” (IMF 2013, 5).

Context: Ireland 2010–2011		
GDP (SAAR, nominal GDP in USD)		\$221.9 billion in 2010
		\$239.0 billion in 2011
GDP per capita (SAAR, nominal GDP in USD)		\$48,664 in 2010
		\$52,183 in 2011
Sovereign credit rating (five-year senior debt)	2010	Moody’s: Baa1
		S&P: A *-
		Fitch: BBB+
	2011	Moody’s: Ba1
		S&P: BBB+ *-
		Fitch: BBB+ *-
Size of banking system		\$353.5 billion in 2010
		\$336.0 billion in 2011
Size of banking system as a % of GDP		159.3% in 2010
		140.6% in 2011
Size of banking system as a % of financial system		Data not available in 2010
		Data not available in 2011
Five-bank concentration of banking system		87.5% in 2010
		86.1% in 2011
Foreign involvement in banking system		36% in 2010
		35% in 2011
Existence of deposit insurance		Yes in 2010
		Yes in 2011
Sources: Bloomberg; World Bank Global Financial Development Database; World Bank Deposit Insurance Dataset.		

Key Design Decisions

- 1. Purpose and Type: The ECB and CBI entered into a bilateral, unidirectional swap arrangement pursuant to which the CBI could request pounds for euros “to facilitate the functioning of financial markets and provide liquidity in GBP.” The BoE and ECB entered into a bilateral, unidirectional swap line to provide pounds (in exchange for euros), which the ECB could provide to Ireland if needed.**

The ECB-CBI swap agreement provided that the CBI could request one or more swap transactions (draws) to sell to (and repurchase from) the ECB euros for pounds. The ECB would simultaneously enter into an agreement to purchase from (and then later resell to) the CBI the euros received from the CBI. All drawdowns under the agreement had to receive the prior approval of the ECB (ECB and CBI 2010). The eurozone finance “ministers concur[ed] with the [European] Commission and the European Central Bank that providing a loan to Ireland is warranted to safeguard financial stability in the euro area and in the European Union as a whole” (BBC News 2010a).

The ECB-CBI swap agreement was bilateral but only unidirectional. In other words, it did not provide for the ECB to request euros from the CBI (ECB 2011b; ECB and CBI 2010).

The BoE-ECB swap agreement described terms under which the Bank of England would provide pounds sterling to the ECB in exchange for euros: “This precautionary measure would enable the ECB to provide sterling liquidity to its counterparties” (BoE 2010). Given this wording and that of the ECB press release of the same date, it seems that the line was prospective and unidirectional, rather than reciprocal, and that the ECB intended that borrowed pounds would be provided to the CBI only and not to other national central banks (ECB 2011b). Once it drew on the line, the ECB could then make the pounds sterling available to the CBI through the ECB-CBI swap arrangement. The CBI could then provide pound liquidity to Irish banks through operations that it administered (Reuters 2010) (ECB 2010a). (See Key Design Decision No. 5, Administration.)

Although the BoE announcement did not mention how any pounds sterling that the ECB drew under the swap line would be used, the ECB’s press release announcing the BoE-ECB agreement stated: “The agreement allows pounds sterling to be made available to the Central Bank of Ireland as a precautionary measure, for the purpose of meeting any temporary liquidity needs of the banking system in that currency” (ECB 2010a). (See Key Design Decision No. 6, Communication.) The agreement, which expired at the end of September 2011, allowed pounds sterling to be made available to the Central Bank of Ireland as a precautionary measure, for the purpose of meeting any temporary liquidity needs of the banking system in that currency. (ECB 2010a). The swap facility was not used in 2010. It is interesting to note that we have not located any ECB or CBI press releases announcing the ECB-CBI swap agreement; it is not clear if this was because it was between the ECB and a euro area national central bank.

Ireland had been particularly hurt by the crisis, and UK banks, particularly the state-backed Royal Bank of Scotland and Lloyds, had lent more than GBP 80 billion to Irish companies and individuals (Treanor 2010). Irish banks had already accessed the ECB's liquidity lines significantly, since other banks refused to lend to them. One-fourth of ECB lending was to Irish banks. The Irish sovereign debt had also been recently downgraded, putting more pressure on the banking system and economy (Irish Examiner 2010; Reuters 2010). The action had significant support among eurozone finance ministers, who issued a statement that said: "[We] concur with the [European] Commission and the ECB that providing a assistance to Ireland is warranted to safeguard financial stability in the EU and in the euro area" (EC 2010).

2. Part of a Package (A): The ECB-CBI and BoE-ECB swaps were part of a package of aid to Ireland provided by the ECB, the UK, and other international authorities.

The ECB-CBI swap was part of significant assistance provided to Ireland by the ECB. Irish banks borrowed euros heavily from the ECB liquidity facilities before the swap was established (CBI 2011). As of November 2010, the ECB had lent to the CBI EUR 140 billion (including emergency liquidity assistance [ELA]⁷) or approximately 85% of Irish GDP. This represented approximately 25% of the ECB's outstanding lending, "an unprecedented level of exposure to a country such as Ireland, whose share in the capital of the ECB was less than one percent" (ECB 2010b).

Additional ECB exposures to Ireland included through the government, which was the issuer and guarantor of some of the securities being used in Eurosystem lending operations. Furthermore, the ECB had a sizeable and increasing exposure to Ireland through TARGET2⁸ (representing the cross-border flows out of Ireland that were recorded as a liability on the CBI's balance sheet).

Prior to entering into the ECB-CBI swap agreement, the president of the ECB and the Irish minister of finance exchanged letters in October and November 2010 wherein the ECB required that the Irish government make certain guarantees and commitments as a precondition to further liquidity support. The Irish government provided a "letter of comfort" in which it also agreed to seek financial support from the European Commission and the IMF (ECB 2010b). (See Key Design Decision No. 12, Balance Sheet Protection.)

The 2010 BoE-ECB swap was implemented as a targeted facility to assist Ireland given the two countries' close trading, financial, and political relationships (BBC News 2010b; Osborne

⁷ ELA is a decision made and administered by the national central bank, but it is closely monitored by the ECB "as it may interfere with the objectives and tasks of the Eurosystem and may contravene the prohibition of monetary financing" (Trichet 2010a, 2). When ELA amounts are significant, the provision of ELA and any subsequent rollover is subject to the non-objection of the ECB's Governing Council (Baudino, Sturluson, and Svoronos 2020, 30). The Irish situation became very controversial, as discussed in Key Design Decision No. 12, Balance Sheet Protection.

⁸ TARGET2 is the Trans-European Automated Real-time Gross Settlement Express Transfer system 2, the ECB's system for settling interbank payments. If the Irish sovereign defaulted, any losses, once collateral had been claimed, would have been shared across the members of the Eurosystem (ECB 2020).

2010). In addition, the UK provided Ireland a GBP 3.2 billion loan as part of an EUR 85 billion bailout from the EU and IMF; Sweden and Denmark also made direct loans to Ireland. Ireland drew down the UK loan between 2010 and 2013. As of March 2021, Ireland had repaid the UK loan and other amounts, but EUR 41 billion in long-term loans were still outstanding to two EU institutions (BBC News 2010b; Campbell 2021; Labanyi 2010; Osborne 2010).

Part of a Package (B): The ECB and the BoE had other currency swap lines in place with other major central banks during the tenure of the ECB-CBI and BoE-ECB swap lines.

The ECB-CBI and BoE-ECB swap lines were adopted in December 2010 when the ECB had other swap lines in place, and several of those lines continued to exist after the ECB-CBI line ended in September 2012 and the BoE-ECB line ended in September 2014.

Notably, the ECB and BoE were part of a coordinated swap arrangement initiated by the Federal Reserve to provide dollar liquidity. On December 12, 2007, the ECB entered into a bilateral, unidirectional swap agreement with the Federal Reserve pursuant to which the ECB could borrow USD 20 billion in exchange for euros to fund US dollar repurchase agreement (repo) loans to its domestic banks. On September 18, 2008, the BoE entered into a similar swap agreement permitting it to borrow a maximum of USD 40 billion in exchange for pounds sterling (BoE 2009; Fed 2008). These agreements were part of a series of swaps that the Fed put in place, ultimately with 14 central banks, to provide dollar liquidity to global markets; the amounts authorized under the agreements were increased over time. In October, the Fed's agreements with four major central banks—BoE, ECB, Bank of Japan (BoJ), and Swiss National Bank (SNB)—were amended to provide whatever amount of US dollar funding those central banks requested. This enabled the ECB and the BoE to conduct operations that lent an unlimited amount of US dollars to their domestic financial institutions (Fed 2013).

The Fed-ECB and Fed-BoE swap agreements were originally authorized for six-month periods but were repeatedly extended, and both expired in February 2010 with the Fed's other US dollar-providing swap lines.

In April 2009, the ECB and BoE each entered into a corresponding swap agreement with the Federal Reserve whereby the ECB would provide the Fed up to EUR 80 billion and the BoE would provide GBP 30 billion in exchange for US dollars so the Fed could provide euro-denominated or sterling-denominated loans to US banks if needed. (The Fed also entered into similar foreign currency swaps with the BoJ and SNB with respect to their currencies.) The Fed did not draw on the ECB or BoE foreign currency-providing lines (Fed n.d.). The parties extended the agreement, and it expired in February 2010 (Fed 2020).

In May 2010, due to the reemergence of strains in US dollar short-term funding markets in Europe, the Fed reestablished temporary dollar liquidity swap lines with five major central banks—the BoE, ECB, SNB, BoJ, and BoC. Each of these swap agreements provided for full allotments at a fixed rate, as had previous agreements between the parties that had been terminated in February 2010 (BoE 2012; Fed 2010).

In November 2011, the six major central banks also entered into a network of temporary bilateral swap arrangements with each other so that liquidity can be provided in each jurisdiction in any of their currencies should market conditions so warrant (BoE 2012; Fed 2011; Fed n.d.). Any central bank drawing on a swap agreement agreed to provide its currency in exchange for the foreign currency requested (Fed 2022).

The network of bilateral agreements was extended several times, and on October 31, 2013, they were converted to a standing arrangement without an express expiration date (Fed n.d.). (See French 2023.) Prior to the COVID-19 pandemic crisis in March 2020, these agreements had been used to lend US dollars and Swiss francs to euro area banks and to lend euros to UK banks (ECB 2022b). The Bank of Japan had also borrowed dollars extensively under these agreements before 2020. (See Hoffner 2023.)

3. Legal Authority: The ECB's Governing Council makes high-level decisions on currency swap agreements. The Bank of England's role in liquidity insurance programs is described in the Financial Services Act (2012).

Foreign currency exchange swaps are a tool used by the Eurosystem for monetary policy purposes (CBI 2022). The Governing Council, the ECB's decision-making body, makes decisions regarding whether to enter into a foreign currency swap agreement (ECB 2022a) (ECB 2021). The ECB normally enters into corresponding swap agreements with national central banks to make foreign currency available to the national central bank for its domestic banks. (See Key Design Decision No. 5, Administration.) Patrick Honohan, then governor, signed the ECB-CBI agreement on behalf of the CBI.

It is well established that the BoE has the authority to engage in foreign currency swaps. The legal basis for the BoE's liquidity insurance facilities is provided by the BoE Act (1998), the Financial Services and Markets Act (2000), the Banking Act (2009), and the Financial Services Act (2012). This legislation is supported with a system of memoranda of understanding (MOUs) and concordats, as well as by the regular exchange of letters between the chancellor of the exchequer and the governor of the BoE (IMF 2016).

The BoE's Market Operations Guide also describes the governance of liquidity programs that execute its monetary and financial stability objectives. That guide describes key terms for short-term, non-sterling—that is, foreign currency—liquidity facilities that the BoE funds through swaps with foreign central banks (BoE 2021).

4. Governance: High-level decisions regarding the foreign currency swap agreement were made by the Court of Directors of the Bank of England and the Governing Council of the European Central Bank.

The ECB's Governing Council is the body that makes the high-level decisions regarding currency swap agreements. Foreign exchange swaps for monetary purposes are part of the Eurosystem framework and apply to the CBI as well (CBI 2022). The ECB reported on the swap line in its annual reports (ECB 2013b). The Governing Council also decided on the conversion into standing arrangements of the regular, nonrestricted swap agreement

between the ECB and BoE that was part of the multilateral network of swaps among six major central banks. (See Key Design Decision No. 2, Part of a Package.)

It would appear that, once approved, the BoE signs the swap agreements and administers the arrangement. Several BoE swap arrangements were announced by the governor of the bank, and a contemporaneous agreement with the Federal Reserve was signed by the BoE executive director of markets on behalf of the governor and company of the BoE (FRBNY and BoE 2010).

5. Administration: Funds drawn under the BoE-ECB swap line would be provided to the CBI to lend to Irish banks through an operation administered by the CBI. The ECB-CBI swap agreement had two administrative factors that were noteworthy.

According to ECB and CBI annual reports, the ECB would use pounds drawn under the BoE-ECB swap to lend pounds to Irish banks through a lending operation that the CBI would administer. In the ECB's usual procedure, it provides euro liquidity directly to euro area banks; however, operations tied to swap arrangements tend to operate differently (CBI 2011; ECB 2011b).

The ECB provided more detail about its US dollar–euro swap line with the Fed, as reported in its 2010 annual report:

Under this arrangement, US dollars were provided by the Federal Reserve to the ECB by means of a temporary swap line, with the aim of offering short-term US dollar funding to Eurosystem counterparties. The ECB simultaneously entered into back-to-back swap transactions with euro area NCBs, which used the resulting funds to conduct US dollar liquidity-providing operations with Eurosystem counterparties in the form of reverse and swap transactions. (ECB 2011b, 226)

The ECB used a similar process with respect to the provision of Swiss franc liquidity under a swap with the Swiss National Bank (ECB 2011b).

The signing of the ECB-CBI swap agreement and the 2010 BoE-ECB swap agreement in close proximity to one another mirrors the ECB's administration of US dollar and Swiss franc liquidity. In the CBI swap arrangement, as in the US dollar–euro swap arrangement with the Fed and the Swiss franc–euro swap arrangement with the Swiss National Bank, it was up to the national central bank, rather than the ECB, to carry out operations with the banks (Reuters 2010). The CBI would be responsible for selecting, reviewing, and monitoring loans to its domestic banks (CBI 2011; ECB 2011b).

We note two administrative elements in the ECB-CBI swap arrangement that appear out of the ordinary. First, while it is customary that upon a draw the parties have to mutually agree to such transaction terms as the amount of the draw, duration, exchange rate, and interest rate, the ECB-CBI agreement included this provision: “all drawdowns must receive the prior approval of the ECB” (ECB and CBI 2010, 1). This wording appears to provide that an additional level of “two-key” approval by the ECB was required, providing the ECB with an additional level of control or “limitation” over the draws. By comparison, neither the ECB's

2010 euro-providing swap agreement with the Fed nor its similar 2014 agreement with the BoE contained such a clause (ECB and BoE 2014; FRBNY and ECB 2010). We are unable to learn any additional information regarding this requirement, but we note that the swap agreement was entered into in the context of Ireland having received significant amounts of liquidity from the ECB, a fact about which the ECB expressed great concern (ECB 2010b). (See Key Design Decision No. 12, Balance Sheet Protection.)

Second, the funds under this agreement appear to have been administered through accounts at the BoE rather than through accounts at each counterparty. The usual practice under a bilateral swap agreement is that the funds are drawn and collateral is deposited and held in an account at the counterparty bank. (See, for example, the 2010 ECB-Fed swap agreement (FRBNY and ECB 2010) and the 2014 ECB-BoE agreement [ECB and BoE 2014].) It appears that under the ECB-CBI agreement, the funds were administered through accounts that the ECB and CBI maintained at the BoE, although we can't be totally certain of this due to partial redactions of the agreement (ECB and CBI 2010). However, administering the swap funds through accounts at the BoE would seem to create an efficiency: instead of pounds being transferred from the BoE to the ECB, and then to the CBI, pounds could be transferred from the BoE to the CBI directly.

The Central Bank of Ireland reported in its 2010 annual report: "As at the date of publication of the accounts [May 20, 2011] one credit institution has availed of this facility" (CBI 2011, 117). However, the CBI did not report any details, and we were not able to confirm this information from the corresponding ECB annual report or other sources. The CBI and ECB made no other public mentions of further use.

The BoE did not report on the administration of the BoE-ECB swap in its annual reporting.

6. Communication: The BoE, ECB, and CBI announced the swap line in press releases with one difference—unlike the BoE, the ECB and CBI described it as limited in purpose and noted that it allowed funds to be provided to the CBI for loans to Irish banks.

The Bank of England announced the 2010 BoE-ECB facility with a brief press release, which stated, "This precautionary measure would enable the ECB to provide sterling liquidity to its counterparties" (BoE 2010). It did not mention Ireland. However, the ECB's release did: "The agreement allows pounds sterling to be made available to the Central Bank of Ireland as a precautionary measure, for the purpose of meeting any temporary liquidity needs of the banking system in that currency" (CBI 2011; ECB 2010a).

The ECB's annual report for that year largely repeated the wording from the press release limiting the swap to the possibility of providing pounds sterling to the CBI: "The agreement, which expires at the end of September 2011, allows pounds sterling to be made available to the Central Bank of Ireland as a precautionary measure, for the purpose of meeting any temporary liquidity needs of the banking system in that currency" (ECB 2011b, 100; ECB 2014, 74).

This description was narrower than the ECB's descriptions of its other swap agreements. For example, its December 2007 agreement with the Federal Reserve—"The Governing Council of the ECB has decided to take joint action with the Federal Reserve by offering US dollar funding to Eurosystem counterparties,"—or a 2008 swap agreement with the Swiss National Bank "to provide Swiss franc funding to Eurosystem counterparties" (ECB 2007; ECB 2014, 66). The BoE-ECB swap agreement is not public. However, annual reports by the CBI and contemporaneous media reports also characterized this swap as a targeted facility (Bennington 2010; CBI 2011; Irish Examiner 2010). In general, it would usually be up to the borrowing central bank, in this case the ECB, to determine how to funnel the liquidity from a swap line to where it is needed in its banking system. So, it would be appropriate for the ECB, not the BoE, to describe which of its counterparties could benefit from the funds even when informal stipulations may exist in the background.

Some media reports posited that the ECB may have wanted to clarify that other member countries that were having difficulties (such as Portugal, Italy, Greece, and Spain) could not look to the facility for funding (Bennington 2010; Reuters 2010). Contemporaneous media reports discussing other UK assistance to Ireland also mentioned some political pushback in the UK against "bailing out" the euro or the EU (BBC News 2010b; Kollewe 2010). While the chancellor argued that assistance to Ireland was justified by the "very specific connections between the UK and Ireland" that made such assistance "completely appropriate," he distanced himself from the possibility of assisting other EU member states (BBC News 2010b).

The ECB-CBI agreement provided that the parties would coordinate the dissemination of any information about the agreement or any draw under the agreement (ECB and CBI 2010). We did not find an announcement of the ECB-CBI agreement or draws taken under it.

7. Eligible Institutions: The ECB intended to distribute the pounds sterling drawn under the 2010 BoE-ECB swap line only to the Central Bank of Ireland for loans to its banking system.

The ECB was the only institution eligible to draw funds under the BoE-ECB swap line (ECB 2010a). The ECB intended that pounds drawn under this agreement would be lent only to the CBI (ECB 2011b). The CBI was the only institution eligible to draw pounds under the ECB-CBI agreement (ECB and CBI 2010).

The ECB is the governing central bank over the eurozone, which issues the euro, a supranational currency that in 2010 was used by 16 countries, including Ireland. The UK was a member of the EU but not of the eurozone and retained its own currency, the pound sterling (Hodson 2011).

The ECB would act as the lender of last resort for the commercial banks of the euro area. Irish banks had borrowed significantly from ECB lending facilities, which provided both euros and US dollars. The ECB-CBI swap line committed the ECB to make funding in pounds sterling available to the CBI, which could then provide an operation for its Irish commercial banks to access such funding (ECB and CBI 2010). The 2010 BoE-ECB swap line would allow

the ECB to acquire pounds sterling in exchange for euros, which it could then make available to the CBI. Under the two agreements, there would be no contractual relationship between the BoE and the CBI, nor any such relationship between the ECB and the borrowing Irish financial institutions with respect to the pounds sterling drawn under the swap line.

8. Size: The ECB-CBI agreement had no size limitations, but amounts had to be preapproved by the ECB. The ECB could request up to GBP 10 billion in exchange for euros from the BoE.

The ECB-CBI agreement did not state an authorized maximum amount of funds that may be drawn, nor did it state a specific limit on the amount of any draw.

The BoE-ECB swap facility permitted the ECB to request up to GBP 10 billion in exchange for euros based on an agreed-upon exchange rate. We have not been able to locate any commentary regarding why the size of the swap line was set at GBP 10 billion. The BoE-ECB swap agreement is not public and, thus, cannot be used to determine if there were any limits on the size of draws that could be made (BoE 2010).

9. Process for Utilizing the Swap Agreement: Under the ECB-CBI and the BoE-ECB swap agreements, the borrowing party would request pounds sterling and deposit the equivalent amount of euros in the counterparty's account.

The ECB-CBI agreement provided that the CBI could request to draw funds under the agreement by providing the ECB with two days' notice of its request. The parties would agree on the following transaction details: amount of the draw, start date of the transaction, exchange rate, and the maturity date of the draw. The agreement did not state an authorized maximum amount of funds that could be drawn, nor did it state a specific limit on the amount of any draw. Once agreed, the terms were to be confirmed by a SWIFT⁹ message (ECB and CBI 2010). All draws were subject to two-key approval, requiring the prior approval of the ECB in addition to the parties agreeing to the usual transaction details. (See Key Design Decision No. 5, Administration.)

The parties would enter into simultaneous repo agreements, and the same exchange rate would apply to the initial purchase of pounds and sale of euros and the subsequent unwind transactions. The interest to be paid was stated in the agreement but redacted. (See Key Design Decision No. 11, Rates and Fees.) The ECB would pay no interest on the euro leg of the transaction (ECB and CBI 2010).

The agreement included rights of setoff in the case of nonperformance by either party. It also provided that in the event that the CBI failed to meet its obligations at the end of any swap transaction, the amounts due (principal and interest) would be converted to an overnight swap and the CBI would be required to immediately deposit additional euros with the ECB. The original termination date was September 30, 2011 (ECB and CBI 2010).

⁹ SWIFT stands for Society for Worldwide Interbank Financial Telecommunication and is a standard, secure method of communication between financial institutions (Wallace 2017).

Under the BoE-ECB agreement, if requested, the BoE would provide the ECB with up to GBP 10 billion in exchange for euros (BoE 2010). The BoE-ECB swap agreement is not public. However, contemporaneous with the 2010 BoE-ECB agreement, the same central banks were party to swap agreements with the Federal Reserve for the provisions of US dollars in exchange for their currencies. Under the terms of those Fed agreements, at the time a party wished to make a draw, the parties would mutually agree on (1) the amount to be borrowed, (2) the date the currency was to be valued, (3) the applicable exchange rate (which applied to both parts of the swap transaction), (4) the maturity date of the draw (not to exceed 88 days), and (5) the interest payable by the borrowing party to the Fed at the unwind. Once agreed, the terms were to be confirmed by a SWIFT message (FRBNY and BoE 2010; FRBNY and ECB 2010).

On the start date of the draw, the borrowing central bank would deposit the required amount of its currency in an account for the benefit of the lending central bank established at its bank. The lending central bank would then deposit the requested amount of its currency in an account at its bank for the benefit of the borrowing central bank. The lending central bank would retain that currency until the expiration of the swap transaction. At the expiration of the swap transaction, the borrowing central bank would return the currency to the lending central bank with an agreed-upon interest and the lending central bank would return the currency it had deposited as security. The lending central bank would pay no interest with respect to the currency returned to the borrowing central bank (FRBNY and BoE 2010; FRBNY and ECB 2010).

According to ECB and CBI annual reports, funds drawn under the swap would be used to lend to Irish banks through an operation administered by the CBI (ECB 2010a). (See Key Design Decision No. 5, Administration.)

10. Downstream Use of Borrowed Funds: The pounds sterling provided under the swap agreements were intended to be used by the CBI to provide liquidity to banks in its jurisdiction if needed.

The intent of the ECB as described in its 2010 annual report seemed to be that once it drew pounds under the ECB-BoE agreement, it could lend those funds to the CBI, pursuant to the ECB-CBI agreement, “as a precautionary measure, for the purpose of meeting any temporary liquidity needs of the banking system in that currency” (ECB 2011b, 100). These needs would presumably be met through the CBI lending pounds to financial markets and banks under operations that it, not the ECB, would administer (ECB and CBI 2010).

11. Duration of Swap Draws: Under the ECB-CBI swap agreement, the parties would mutually agree on the duration of a draw, which appears to be linked to BoE Monetary Policy Committee bank rate announcement dates. No public sources describe the maturity of draws under either swap facility.

Under the ECB-CBI swap agreement, the parties would mutually agree on the duration of a draw, and each maturity date “shall not extend beyond the next day following the [start date of the draw] on which of the Bank of England’s Monetary Policy Committee is scheduled to

determine [redacted] . . . or as otherwise agreed by the ECB” (ECB and CBI 2010, 2). This likely referred to the dates on which the Monetary Policy Committee announced the bank rate (the interest rate it charges banks to borrow), which it determines and publishes eight times a year (BoE 2022a). The intended announcement dates are published for the current and next year and occur roughly six to eight weeks apart (BoE 2022b).

We have found no public information regarding any draws under this facility. However, the CBI reported that one party utilized the BoE-ECB swap the facility but did not report the maturity of the draw or other details (CBI 2011). The BoE-ECB swap agreement is not public.

12. Rates and Fees: Although we cannot be definitive, it appears that the ECB-CBI swap agreement charged interest at the UK bank rate for draws of pounds. We found no information relating to rates and fees charged under the BoE-ECB swap agreement.

The ECB-CBI agreement provided for the rate of interest to be paid on the GBP leg of the transaction. Although the specific rate has been redacted, available text is revealing:

Interest will be payable by the CBI on the GBP part of each Transaction at the prevailing [redacted] for purchases by the CBI of GBP for the period corresponding to the period between the [start date] and the Maturity Date of the Transaction. The [redacted] is determined by the Bank of England’s Monetary Policy Committee and published by the Bank of England from time to time. (ECB and CBI 2010, 2)

This likely referred to the bank rate, which is the key interest rate that the BoE Monetary Policy Committee sets and publishes eight times a year (BoE 2022b).

Interest was calculated on a 365-day basis for the period that the draw was outstanding. The agreement also stated that the ECB would not pay any interest with respect to the euro part of a draw (ECB and CBI 2010).

We have not been able to locate the BoE-ECB swap agreement or any information regarding the exchange rates, interest rate, or fees applied to the swap. However, for information, the 2014 BoE-ECB swap agreement provided that the parties would agree on the interest rate when a draw was initiated (ECB and BoE 2014).

13. Balance Sheet Protection: The ECB, the lending central bank, took extra measures to protect its exposure under the ECB-CBI agreement and to Ireland generally. There is no public information about measures the BoE took to protect its exposures as the lending central bank under the 2010 BoE-ECB agreement.

The ECB-CBI swap was part of significant assistance provided to Ireland by the ECB from 2007 onward. Irish banks borrowed euros heavily from the ECB liquidity facilities before the swap line was established (CBI 2011). As of November 2010, the ECB had lent to the CBI EUR 140 billion (including ELA) or approximately 85% of Irish GDP. This represented approximately 25% of the ECB’s outstanding lending, to Irish banks, “an unprecedented level of exposure to a country such as Ireland, whose share in the capital of the ECB was less than

one percent” (ECB 2010b). The ECB also had additional exposures to Ireland because its government was the issuer or guarantor of securities used as collateral in Eurosystem lending operations. (See also Key Design Decision No. 2, Part of a Package.)

As a result, the ECB and Irish government exchanged a series of letters (ECB 2010b). On October 15, 2010, the then-president of the ECB, Jean-Claude Trichet, wrote to the Irish minister of finance expressing concern regarding “the extraordinarily large provision of liquidity by the Eurosystem to Irish banks in recent weeks” (Trichet 2010a, 1). Trichet issued a follow-up letter on November 19, 2010, stating further that “the exposure of the Eurosystem and the Central Bank of Ireland vis-à-vis Irish financial institutions has risen significantly over the past few months to levels that we consider with great concern. Recent developments can only add to these concerns” (Trichet 2010b, 1). The ECB requested that the Irish government provide it a “letter of comfort” covering four points as a condition of further liquidity support:

- 1) The Irish Government shall send a request for financial support to the Eurogroup;
- 2) The request shall include the commitment to undertake decisive actions in the areas of fiscal consolidation, structural reforms and financial sector restructuring, in agreement with the European Commission, the [IMF] and the ECB;
- 3) The plan for the restructuring of the Irish financial sector shall include the provision of the necessary capital to those Irish banks needing it and will be funded by the financial resources provided at the European and international level to the Irish government as well as by the financial means currently available to the Irish government, including existing cash reserves of the Irish government;
- 4) The repayment of the funds provided in the form of ELA shall be fully guaranteed by the Irish Government, which would ensure the payment of immediate compensation to the Central Bank of Ireland in the event of missed payments on the side of the recipient institutions. (Trichet 2010b, 1–2).

There is no public information about measures the BoE took to protect its exposures under the BoE-ECB swap agreement. Under typical foreign currency swaps between central banks, the loaned currency (in this case pounds sterling) would be secured by the collateral received (euros) on the lending central bank’s balance sheet. Should the ECB not be able to return the borrowed pounds sterling at the maturity of the swap, the BoE could offset the ECB’s obligations against euros that it held euros in the ECB’s account (Bahaj and Reis 2018). (See ECB and CBI [2010, 4] for an example of a setoff clause.)

Because the exchange rate would be fixed in advance, neither the lending bank (in this case, the BoE) nor the borrowing bank (the ECB) would experience any exchange rate risk. The lending bank would hold the currency it received (in this case euros) as uninvested collateral and the same amount of pounds sterling it provided would be returned to it at the end of the swap. Similarly, because the interest rate would be fixed at the beginning of the swap draw, neither the lending nor borrowing central bank would experience any interest rate risk (Bahaj and Reis 2018).

14. Other Restrictions: The ECB intended that pounds sterling it drew under the BoE-ECB swap line would be lent only to the CBI for it to lend to Irish banks.

ECB and CBI public reports and statements indicate that funds drawn under the BoE-ECB swap line would be provided only to the CBI to be used to provide euro liquidity to banks in Ireland (ECB 2010a). There is no public information about other restrictions imposed by the ECB or BoE on the borrowing central banks under the ECB-CBI or BoE-ECB agreement. (But see Key Design Decision No. 9, Process for Utilizing the Swap Agreement.)

15. Other Options: We found no evidence that the ECB or the BoE considered other options before implementing the swap lines.

We found no evidence that the ECB or the BoE considered other options in lieu of the swap lines. However, the BoE did discuss its other forms of direct (loan) and indirect assistance (through the IMF and EU) to Ireland contemporaneously with implementing the swaps (BBC News 2010b).

16. Exit Strategy: The ECB-CBI and BoE-ECB swap agreements originally carried nine-month terms but were extended before expiring.

The ECB-CBI swap line was originally scheduled to expire on September 30, 2011, but in August 2011, it was extended to September 2012, when it presumably expired (ECB 2011c).

The BoE-ECB swap line was originally scheduled to terminate on September 30, 2011, but was renewed three times and eventually expired on September 30, 2014 (ECB 2014) (ECB 2012; ECB 2013a). The press releases announcing the extensions did not state a reason (ECB 2012). The Central Bank of Ireland reported in its 2010 annual report, with respect to the swap line, that “one credit institution has availed of this facility” as of May 20, 2011 (CBI 2011, 117). However, the corresponding ECB annual report did not mention usage of the facility, and no other public documents provide other evidence of further use.

References and Key Program Documents

Implementation Documents

(BoE 2021) Bank of England (BoE). 2021. Bank of England Market Operations Guide: Our Tools.

Overview of the Sterling Monetary Framework.

<https://ypfs.som.yale.edu/node/17564>

(ECB and BoE 2014) European Central Bank, and Bank of England (ECB and BoE). 2014. "Euro-British Pounds Swap Agreement Dated as of 16 January 2014," January 16, 2014.

Swap agreement between the ECB and BoE.

<https://ypfs.som.yale.edu/node/22797/>

(ECB and CBI 2010) European Central Bank, and Central Bank of Ireland (ECB and CBI). 2010. "Euro Pound Swap Facility Agreement," October 12, 2010.

Swap agreement between the ECB and CBI dated December 10, 2010.

<https://ypfs.som.yale.edu/node/22727>

(ECB and CBI 2011c) European Central Bank and Central Bank of Ireland (ECB and CBI). 2011c. "Amendment Dated as of 22 August 2011 to the Euro Pound Swap Facility Agreement Dated as of 10 December 2010," August 22, 2011.

Amendment to the swap agreement between the ECB and CBI dated December 10, 2010, extending it to September 2012.

<https://ypfs.som.yale.edu/node/22849>

Legal/Regulatory Guidance

(ECB 2021) European Central Bank (ECB). 2021. "Governing Council." June 19, 2021.

Web page detailing the responsibilities and members of the ECB's Governing Council.

<https://ypfs.som.yale.edu/node/22220>

Media Stories

(BBC News 2010a) BBC News. 2010a. "Irish Republic 85bn Euro Bail-out Agreed," November 28, 2010.

News article that writes about the bail-out for Ireland.

<https://ypfs.som.yale.edu/node/22716>

(BBC News 2010b) BBC News. 2010b. "Irish Republic Bail-out in UK Interest - George Osborne," November 22, 2010.

News article that writes about the bail-out program for Ireland.

ypfs.som.yale.edu/node/21873/

(Bennington 2010) Bennington, Ash. 2010. "Bank of England Swap Line: Slicing & Dicing the PIIGS." CNBC, December 17, 2010.

News article about the BoE-ECB swap.

<https://ypfs.som.yale.edu/node/21874>

(Campbell 2021) Campbell, John. 2021. "Irish Make Final Repayment on UK Emergency 'bailout' Loan." BBC News, March 26, 2021.

News article regarding Ireland repaying UK emergency loan.

<https://ypfs.som.yale.edu/node/21877>

(Dow Jones News 2008) Dow Jones News. 2008. "DJ Update: Irish Government Moves to Safeguard Banking System," September 30, 2008.

News article that announces blanket guarantee on all deposits, covered bonds, senior debt and dated subordinated debt of the four main banks.

<https://ypfs.som.yale.edu/node/21879/>

(Irish Examiner 2010) Irish Examiner. 2010. "ECB and Bank of England Swap Line," December 18, 2010.

News article discussing the swap line between the ECB and the BoE.

<https://ypfs.som.yale.edu/node/21890/>

(Kollewe 2010) Kollewe, Julia. 2010. "Ireland Bailout: UK to Lend £7bn." *The Guardian*, November 22, 2010.

News article that details the bailout of Ireland.

<https://ypfs.som.yale.edu/node/21892>

(Labanyi 2010) Labanyi, David. 2010. "Ireland to Receive €85 Billion Bailout at 5.8% Interest Rate." *The Irish Times*, November 28, 2010.

News article detailing the bailout aid received by Ireland.

<https://ypfs.som.yale.edu/node/21893>

(McDonald and Treanor 2010) McDonald, Henry, and Jill Treanor. 2010. "Ireland Asks for €90bn EU Bailout." *The Guardian*, November 21, 2010.

News article discussing the bailout program.

<https://ypfs.som.yale.edu/node/21894>

(Murray-Brown and Dennis 2008) Murray-Brown, John, and Neil Dennis. 2008. "Ireland Guarantees Six Banks' Deposits." *Financial Times*, September 30, 2008.

News article covering the guarantee program initiated by Ireland.

<https://ypfs.som.yale.edu/node/21896/>

(Neuger and Kennedy 2010) Neuger, James G., and Simon Kennedy. 2010. "Ireland Wins EU85 Billion; Germany Drops Bond Threat." *Bloomberg*, November 29, 2010.

News article discussing details of the package given to Ireland.

<https://ypfs.som.yale.edu/node/21875>

(Reuters 2010) Reuters. 2010. "ECB Signs Sterling Swap Deal to Aid Irish Banks," December 17, 2010.

News article detailing the swap arrangement between the ECB and BoE.

<https://ypfs.som.yale.edu/node/21899/>

(Reuters Staff 2008) Reuters Staff. 2008. "Ireland Raises State Guarantee on Irish Deposits." Reuters, September 20, 2008.

Article discussing Ireland's decision to increase its depositor coverage to EUR 100,000.

<https://ypfs.som.yale.edu/node/21865/>

(Saltmarsh 2010) Saltmarsh, Matthew. 2010. "Citing Weak Banks, Moody's Downgrades Ireland's Debt." *New York Times*, July 20, 2010.

News story describing rating agency downgrades of Ireland's debt.

<https://ypfs.som.yale.edu/node/21900/>

(Treanor 2010) Treanor, Jill. 2010. "Ireland: AIB Crisis Shows UK Must Help Bailout to Save Its Own Banks." *The Guardian*, November 18, 2010.

News article that sheds light on the severity of the Irish crisis.

<https://ypfs.som.yale.edu/node/21902/>

Press Releases/Announcements

(BoE 2010) Bank of England (BoE). 2010. "European Central Bank Swap Line Agreement." Press release, December 17, 2010.

Press release announcing the swap with the ECB.

<https://ypfs.som.yale.edu/node/21870/>

(BoE 2012) Bank of England (BoE). 2012. "Extension of Existing Central Bank Swap Arrangements." Press release, December 13, 2012.

Press release announcing the extension of the existing swap arrangements with the ECB.

<https://ypfs.som.yale.edu/node/21871/>

(ECB 2007) European Central Bank (ECB). 2007. "Measures Designed to Address Elevated Pressures in Short-Term Funding Markets." Press release, December 12, 2007.

Press release announcing ECB measures to address elevated pressures in short-term funding.

<https://ypfs.som.yale.edu/node/21886/>

(ECB 2010a) European Central Bank (ECB). 2010a. "ECB Signs Swap Facility Agreement with the Bank of England." Press release, December 17, 2010.

Press release announcing the swap line with the BoE.

<https://ypfs.som.yale.edu/node/21880/>

(ECB 2011a) European Central Bank (ECB). 2011a. "Prolongation of Swap Facility Agreement with the Bank of England." Press release, August 25, 2011.

Press release announcing extension of the swap with the BoE.

<https://ypfs.som.yale.edu/node/23021/>

(ECB 2012) European Central Bank (ECB). 2012. "ECB Extends the Swap Facility Agreement with the Bank of England." Press release, September 12, 2012.
Press release announcing the extension of the swap line with the BoE.
<https://ypfs.som.yale.edu/node/21881/>

(ECB 2013a) European Central Bank (ECB). 2013a. "ECB Extends the Swap Facility with the Bank of England." Press release, September 16, 2013.
Press release announcing extension of the BoE swap line.
<https://ypfs.som.yale.edu/node/21882/>

(ECB 2013b) European Central Bank (ECB). 2013b. "ECB Establishes Standing Swap Arrangements with Other Central Banks." Press release, October 31, 2013.
Press release announcing that the multilateral network of swaps between six major central banks has been converted to standing facility with no expiration date.
<https://ypfs.som.yale.edu/node/21883>

(ECB 2022a) European Central Bank. 2022a. "ECB Extends Liquidity Lines with Non-Euro Area Central Banks until 15 January 2024." Press release, December 15, 2022.
Press release announcing the extension of liquidity lines to January 2024.
<https://ypfs.som.yale.edu/node/22850>

(EC 2010) European Commission (EC). 2010. "Statement by the Eurogroup and ECOFIN Ministers on Ireland." Press release, November 22, 2010.
Statement by the Eurogroup and ECOFIN Ministers on Ireland on the latter's request for financial assistance.
<https://ypfs.som.yale.edu/node/22901/>

(Fed 2008) Federal Reserve Board (Fed). 2008. "Federal Reserve and Other Central Banks Announce Further Coordinated Actions to Expand Significantly the Capacity to Provide U.S. Dollar Liquidity." Press release, September 29, 2008, September 29, 2008.
Press release announcing coordinated central bank actions in the wake of the collapse of Lehman Brothers, including a USD 330 billion expansion of swap lines.
<https://ypfs.som.yale.edu/node/20177>

(Fed 2010) Federal Reserve Board (Fed). 2010. "FOMC Statement: Federal Reserve, European Central Bank, Bank of Canada, Bank of England, and Swiss National Bank Announce Reestablishment of Temporary U.S. Dollar Liquidity Swap Facilities." Press release, May 9, 2010.
Press release announcing that the Fed and five major central banks are reestablishing a temporary multilateral network of swaps.
<https://ypfs.som.yale.edu/node/21876>

(Fed 2011) Federal Reserve Board (Fed). 2011. "Coordinated Central Bank Action to Address Pressures in Global Money Markets." Press release, November 30, 2011.
Press release announcing coordinated action by six major central banks regarding a multilateral network of swaps to address pressures in global money markets.
<https://ypfs.som.yale.edu/node/22412>

(Fed 2013) Federal Reserve Board (Fed). 2013. "Federal Reserve and Other Central Banks Convert Temporary Bilateral Liquidity Swap Arrangements to Standing Arrangements." Press release, October 31, 2013.

Press release announcing the conversion of temporary swap lines to standing swap facilities.
<https://ypfs.som.yale.edu/node/22231/>

(Moody's 2010) Moody's Investors Service (Moody's). 2010. "Moody's Downgrades Ireland to Baa1 from Aa2; Outlook Negative." Press release, December 17, 2010.

Rating Action downgrading Irish sovereign debt.
<https://ypfs.som.yale.edu/node/21895/>

(Osborne 2010) Osborne, George. 2010. "Statement by the Chancellor of the Exchequer, Rt Hon George Osborne MP, Statement on Financial Assistance for Ireland." Speech delivered at the House of Commons on November 22, 2010, November 22, 2010.

Statement by the Chancellor of the Exchequer on support for Ireland.
<https://ypfs.som.yale.edu/node/21897>

Reports/Assessments

(BoE 2009) Bank of England (BoE). 2009. Annual Report 2009.

Financial statements for the year as of February 28, 2009.
<https://ypfs.som.yale.edu/node/14974>

(CBI 2011) Central Bank of Ireland (CBI). 2011. Central Bank of Ireland Annual Report 2010.

Yearly report of activity by the CBI.
<https://ypfs.som.yale.edu/node/21867>

(ECB 2011b) European Central Bank (ECB). 2011b. "Annual Report 2010," 2011.

Yearly report of activity by the ECB.
<https://ypfs.som.yale.edu/node/21868/>

(IMF 2013) International Monetary Fund (IMF). 2013. "Ireland: Eleventh Review Under the Extended Arrangement." IMF Country Report No. 13/305, October 2013.

IMF Country Report that assesses the economic conditions in Ireland.
<https://ypfs.som.yale.edu/node/21869/>

Key Academic Papers

(Bahaj and Reis 2018) Bahaj, Saleem, and Ricardo Reis. 2018. "Central Bank Swap Lines," September 25, 2018.

Paper discussing the ECB's experience with swap lines since 2007.
<https://ypfs.som.yale.edu/node/21493>

(Baudino, Murphy, and Svoronos 2020) Baudino, Patrizia, Diarmuid Murphy, and Jean-Philippe Svoronos. 2020. "The Banking Crisis in Ireland." FSI Crisis Management Series No. 2, October 2020.

Paper focusing on the policy response to the 2008 banking crisis in Ireland.

<https://ypfs.som.yale.edu/node/17398>

(Baudino, Sturluson, and Svoronos 2020) Baudino, Patrizia, Jon Thor Sturluson, and Jean-Philippe Svoronos. 2020. "The Banking Crisis in Iceland." Bank for International Settlements, Financial Stability Institute Crisis Management Series No. 1, March 2020.

BIS paper discussing Iceland's crisis and responses.

<https://ypfs.som.yale.edu/node/20306/>

(BoE 2022b) Bank of England (BoE). 2022b. "Monetary Policy Committee Dates for 2023 and 2024."

Meeting minutes and the quarterly Monetary Policy Report for 2023 and 2024.

<https://www.bankofengland.co.uk/monetary-policy/upcoming-mpc-dates>

(ECB 2014) European Central Bank (ECB). 2014. "Experience with Foreign Currency Liquidity-Providing Central Banks Swaps." Monthly Bulletin, August 2014.

Article discussing the ECB's experience with swap lines since 2007.

<https://ypfs.som.yale.edu/node/21884>

(French 2023) French, Jack. 2023. "United States: Central Bank Swaps to Five Countries, 2010–2011." *Journal of Financial Crises* 5, no. 1.

Case study discussing the Fed's swaps to five countries.

<https://elischolar.library.yale.edu/journal-of-financial-crises/vol5/iss1/19>

(Hodson 2011) Hodson, Dermot. 2011. "The EU Economy: The Eurozone in 2010." *Journal of Common Market Studies* (JCMS), March 8, 2011.

Journal article discussing the developments in the eurozone area in 2010.

<https://ypfs.som.yale.edu/node/22713>

(Hoffner 2023) Hoffner, Benjamin. 2023. "United States: Central Bank Swaps to 14 Countries, 2020." *Journal of Financial Crises* 5, no. 1.

Case study discussing the Fed's swaps to 14 countries during the COVID-19 crisis.

<https://elischolar.library.yale.edu/journal-of-financial-crises/vol5/iss1/21>

(IMF 2016) International Monetary Fund (IMF). 2016. "United Kingdom-Financial Sector Assessment Program," June 2016.

IMF FSAP report on the UK.

<https://ypfs.som.yale.edu/node/22851>

(Thorhallsson and Kirby 2012) Thorhallsson, Baldur, and Peadar Kirby. 2012. "Financial Crises in Iceland and Ireland: Does European Union and Euro Membership Matter?" *Journal of Common Market Studies* (JCMS), 2012.

Paper discussing the value of EU and eurozone membership to Ireland in light of its crisis.

<https://ypfs.som.yale.edu/node/21901>

Other

(BoE 2022a) Bank of England (BoE). 2022a. "Monetary Policy."

Web page describing how the BoE conducts its monetary policy.

<https://ypfs.som.yale.edu/node/18283>

(CBI 2022) Central Bank of Ireland (CBI). 2022. Documentation on Monetary Policy Instruments and Procedures.

Manual regarding monetary policy instruments and procedures.

<https://ypfs.som.yale.edu/node/22717>

(ECB 2010b) European Central Bank (ECB). 2010b. "Irish Letters." Letter, November 2010, 2010.

A series of letters between the ECB and the Irish government regarding concerns over aid to Ireland and terms of further assistance.

<https://ypfs.som.yale.edu/node/22457/>

(ECB 2020) European Central Bank (ECB). 2020. "What Are TARGET2 Balances?," December 8, 2020.

Document that describes TARGET2 balances.

<https://ypfs.som.yale.edu/node/22715>

(ECB 2022b) European Central Bank (ECB). 2022b. "What Are Currency Swap Lines?" Web page, May 18, 2022.

ECB webpage discussing foreign currency swap lines.

<https://ypfs.som.yale.edu/node/21885>

(Fed n.d.) Federal Reserve Board (Fed). n.d. "Central Bank Liquidity Swaps." Accessed July 27, 2022.

Federal Reserve Board web page describing central bank swap lines.

<https://ypfs.som.yale.edu/node/21501>

(Fed 2020) Federal Reserve Board (Fed). 2020. "Central Bank Liquidity Swap Lines." Web page, April 15, 2020.

Fed Reserve Board web page describing central bank swap lines since 2007.

<https://ypfs.som.yale.edu/node/22456>

(Fed 2022) Federal Reserve Board (Fed). 2022. "Federal Reserve Board - Frequently Asked Questions: U.S. Dollar and Foreign Currency Liquidity Swaps." Web page, 2022.

FAQ describing Fed swap lines.

<https://ypfs.som.yale.edu/node/21889>

(FRBNY and ECB 2010) Federal Reserve Bank of New York and European Central Bank (FRBNY and ECB). 2010. "U.S. Dollar-Euro Swap Agreement Dated May 10, 2010," May 10, 2010.

Swap agreement between the Fed and ECB dated May 10, 2010.

<https://ypfs.som.yale.edu/node/21887>

(FRBNY and BoE 2010) Federal Reserve Bank of New York and Bank of England (FRBNY and BoE). 2010. "U.S. Dollar-GBP Swap Agreement Dated May 10, 2010," May 10, 2010. *Swap agreement between the Fed and BoE dated May 10, 2010*.
<https://ypfs.som.yale.edu/node/21888/>

(Trichet 2010a) Trichet, Jean-Claude. 2010a. "Letter from Trichet to Lenihan Dated October 15, 2010," October 15, 2010. *Letter from then-ECB President Trichet to then-Finance Minister of Ireland Lenihan regarding concerns over aid to Ireland*.
<https://ypfs.som.yale.edu/node/22837>

(Trichet 2010b) Trichet, Jean-Claude. 2010b. "Letter from ECB President to Ireland Finance Minister Dated November 19, 2010," November 19, 2010. *Letter from then-ECB President Trichet to then-Finance Minister of Ireland Lenihan regarding terms and conditions for future aid*.
<https://ypfs.som.yale.edu/node/22838>

(Wallace 2017) Wallace, K. Lynn. 2017. "What Is SWIFT Messaging?" Bizfluent, September 9, 2017. *Article discussing SWIFT messaging*.
<https://ypfs.som.yale.edu/node/21903>

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